MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

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References:

1. R. Gaonkar, Microprocessor Architecture Programming, Penram International Publishing
2. Hall D.V., microprocessor & Interfacing-Programming & hardware, TMH, 2008

Faculty HOD Dean Academics

UNIT-1 (Introduction and Architecture of 8085)

Introduction & Evolution of Microprocessor

Lecture No.-1

A Microprocessor is an important part of computer architecture without which not possible to perform anything on computer. It is a programmable device that takes in input perform some arithmetic and logical operations over it and produce desired output

In simple words, a Microprocessor is a digital device on a chip which can fetch instruction from memory, decode and execute them and give results.

Evolution of Microprocessor

Transistor was invented in 1948 (23 December 1947 in Bell lab).

IC was invented in 1958 (Fair Child Semiconductors) By Texas Instruments J Kilby.

The first microprocessor was invented by INTEL(INTegratedELectronics).

Name	Year of Invention	Clock Speed	Size of MP
INTEL 4004/4040	1971	INTEL 4004/4040	4 Bit
8008	1972	500 kHz	8 Bit
8080	1974	2 MHz	8 Bit
8085	1976 (16-bit address bus)	3 MHz	8 Bit
8086	1978 (16-bit data bus and 20-bit address bus)	4.77 MHz, 8 MHz, 10 MHz	16 Bit
8088	1979 (cheaper version of 8086 and 8-bit external bus)		16 Bit
80186/80188	1982 (80188 cheaper version of 80186, and additional components like interrupt controller, clock generator, local bus controller, counters)	6 MHz	16 Bit
80286	1982 (data bus 16bit and address bus 24 bit)	8 MHz	16 Bit
INTEL 80386	1986 ,data bus 32-bit address bus 32 bit	16 MHz – 33 MHz	32 Bit
INTEL 80486	1986 (other versions 80486DX, 80486SX, 80486DX2, 80486DX4)	16 MHz – 100 MHz	32 Bit
PENTIUM	1993	66 MHz	32 Bit

INTEL core 2	2006 (other versions core2 duo, core2 quad, core2 extreme)	1.2 GHz to 3 GHz	64 Bit
i3, i5, i7	2007, 2009, 2010	2.2GHz – 3.3GHz, 2.4GHz – 3.6GHz, 2.93GHz – 3.33GHz	64 Bit

Features of Microprocessor :

Clock speed: One of the earliest features of microprocessors was the clock speed, which refers to the speed at which the processor can execute instructions. Over time, clock speeds have increased, with modern processors capable of speeds in the billions of cycles per second (GHz).

Instruction set architecture: Microprocessors have evolved to support different instruction set architectures, including CISC (complex instruction set computer) and RISC (reduced instruction set computer), which affect the efficiency and complexity of processing.

Cache memory: Microprocessors now include a cache memory, which is a small amount of high-speed memory that stores frequently used data for quicker access.

Multi-core processors: Modern microprocessors have multiple cores, allowing for multiple tasks to be executed simultaneously, increasing performance and multitasking capabilities.

Virtualization: Microprocessors now support virtualization, which enables multiple operating systems to run on the same physical hardware.

Power management: Modern processors include power management features, which reduce power consumption and improve energy efficiency.

Graphics processing: Many modern microprocessors include integrated graphics processing units (GPUs), which allow for faster and more efficient handling of graphics-intensive tasks.

Security features: Microprocessors now include security features, such as hardware-level encryption and secure boot, to protect against malware and hacking.

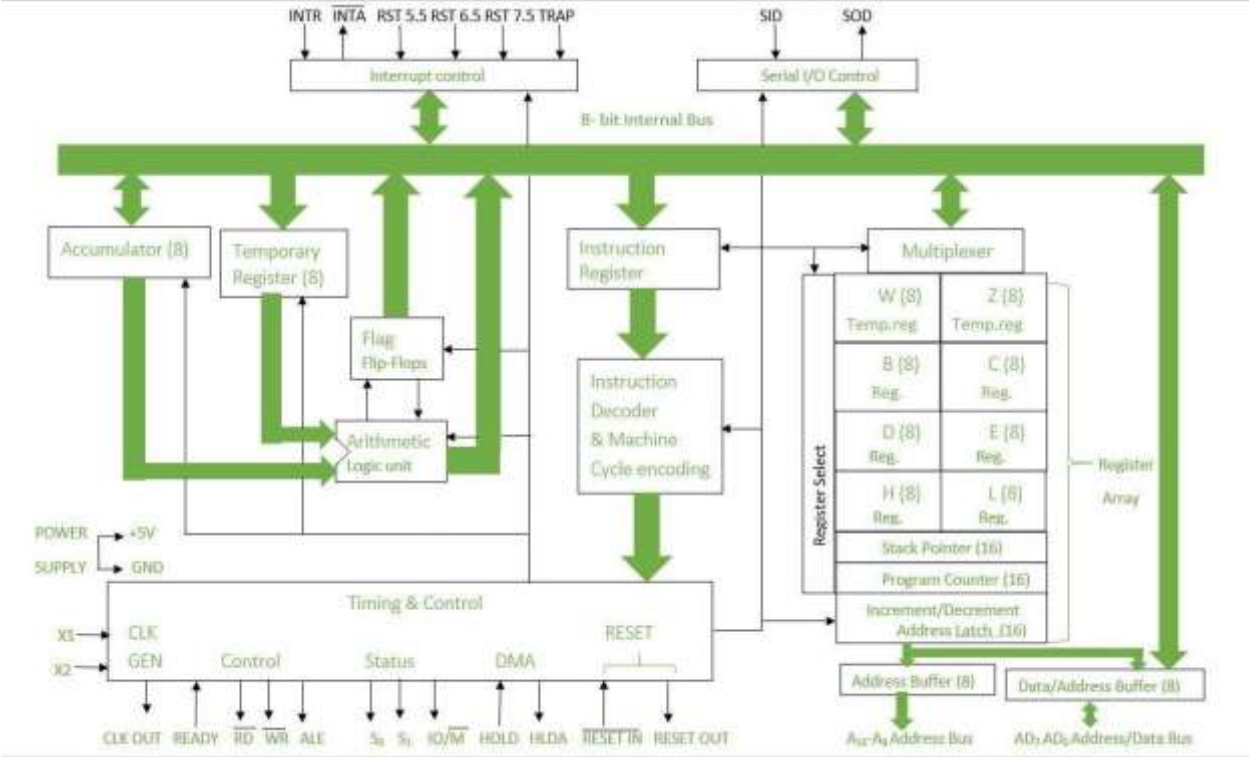
Internet connectivity: Microprocessors now include built-in networking capabilities, such as Wi-Fi and Ethernet, which allow for seamless internet connectivity.

Machine learning capabilities: Some modern microprocessors include specialized processing units for machine learning and artificial intelligence tasks, allowing for faster and more efficient processing of these tasks

8085 MP Architecture

Lecture-2

8085 Microprocessor Architecture & Operations



It is an 8-bit microprocessor designed by Intel in 1977 using NMOS technology.

It has the following configuration −

- 8-bit data bus
- 16-bit address bus, which can address upto 64KB
- A 16-bit program counter
- A 16-bit stack pointer
- Six 8-bit registers arranged in pairs: BC, DE, HL
- Requires +5V supply to operate at 3.2 MHZ single phase clock

It is used in washing machines, microwave ovens, mobile phones, etc.

8085 Microprocessor – Functional Units

8085 consists of the following functional units :

Accumulator

It is an 8-bit register used to perform arithmetic, logical, I/O & LOAD/STORE operations. It is connected to internal data bus & ALU.

Arithmetic and logic unit

As the name suggests, it performs arithmetic and logical operations like Addition, Subtraction, AND, OR, etc. on 8-bit data.

General purpose register

There are 6 general purpose registers in 8085 processor, i.e. B, C, D, E, H& L. Each register can hold 8-bit data.

These registers can work in pair to hold 16-bit data and their pairing combination is like B-C, D-E & H-L.

Program counter

It is a 16-bit register used to store the memory address location of the next instruction to be executed. Microprocessor increments the program whenever an instruction is being executed, so that the program counter points to the memory address of the next instruction that is going to be executed.

Stack pointer

It is also a 16-bit register works like stack, which is always incremented/decremented by 2 during push & pop operations.

Temporary register

It is an 8-bit register, which holds the temporary data of arithmetic and logical operations.

Flag register

It is an 8-bit register having five 1-bit flip-flops, which holds either 0 or 1 depending upon the result stored in the accumulator.

These are the set of 5 flip-flops −

- Sign (S)
- Zero (Z)
- Auxiliary Carry (AC)
- Parity (P)
- Carry (C)

Instruction register and decoder

It is an 8-bit register. When an instruction is fetched from memory then it is stored in the Instruction register. Instruction decoder decodes the information present in the Instruction register.

Timing and control unit

It provides timing and control signal to the microprocessor to perform operations. Following are the timing and control signals, which control external and internal circuits −

- Control Signals: READY, RD', WR', ALE
- Status Signals: S0, S1, IO/M'
- DMA Signals: HOLD, HLDA
- RESET Signals: RESET IN, RESET OUT

Interrupt control

As the name suggests it controls the interrupts during a process. When a microprocessor is executing a main program and whenever an interrupt occurs, the microprocessor shifts the control from the main program to process the incoming request. After the request is completed, the control goes back to the main program.

There are 5 interrupt signals in 8085 microprocessor: INTR, RST 7.5, RST 6.5, RST 5.5, TRAP.

Serial Input/output control

It controls the serial data communication by using these two instructions: SID (Serial input data) and SOD (Serial output data).

Address buffer and address-data buffer

The content stored in the stack pointer and program counter is loaded into the address buffer and address-data buffer to communicate with the CPU. The memory and I/O chips are connected to these buses; the CPU can exchange the desired data with the memory and I/O chips.

Address bus and data bus

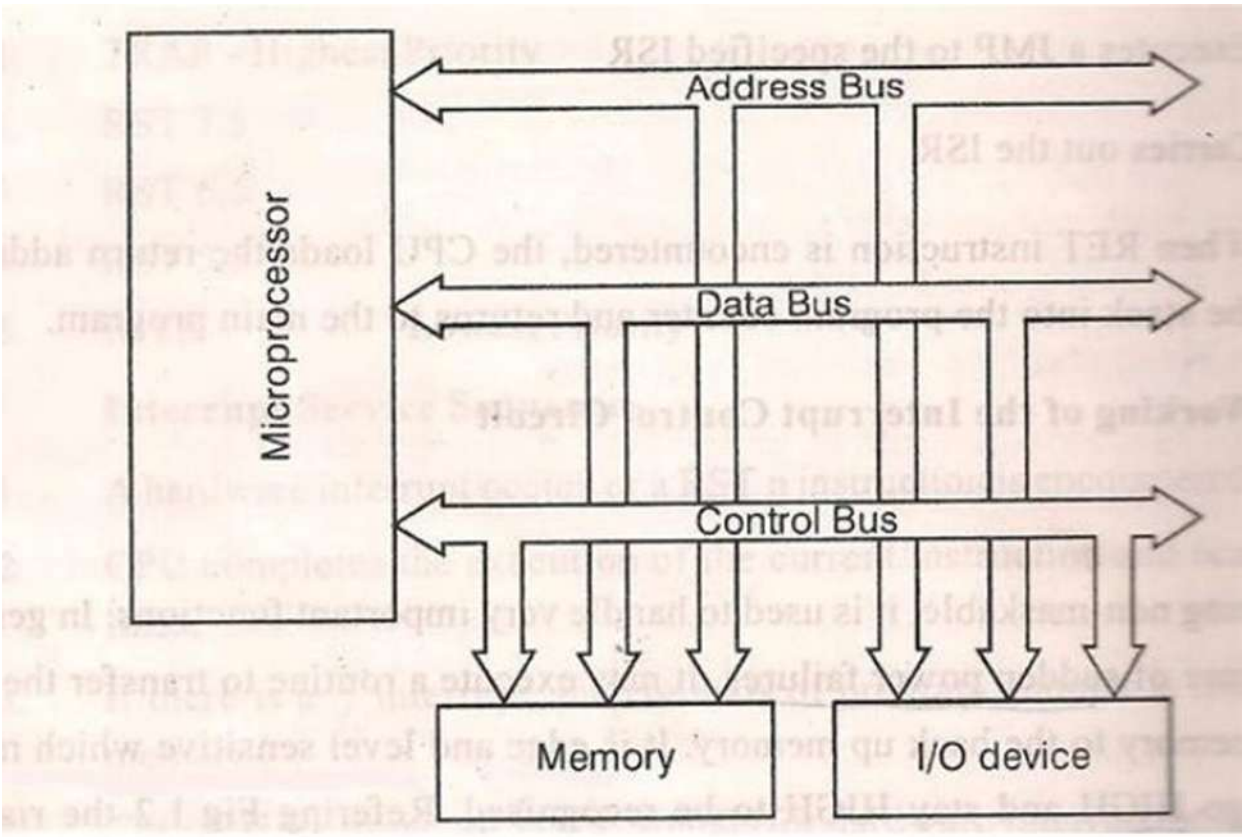
Data bus carries the data to be stored. It is bidirectional, whereas address bus carries the location to where it should be stored and it is unidirectional. It is used to transfer the data & Address I/O devices.

Memory and I/O Operation

Lecture-3

Interfacing the 8085 microprocessor:

A microprocessor is the CPU of a computer. A microprocessor can perform some operation on a data and give the output. But to perform the operation we need an input to enter the data and an output to display the results of the operation. So we are using a keyboard and monitor as Input and output along with the processor. Microprocessors engineering involves a lot of other concepts and we also interface memory elements like ROM, EPROM to access the memory.



Interfacing Types:

There are two types of interfacing in context of the 8085 Microprocessor. Memory Interfacing & I/O Interfacing.

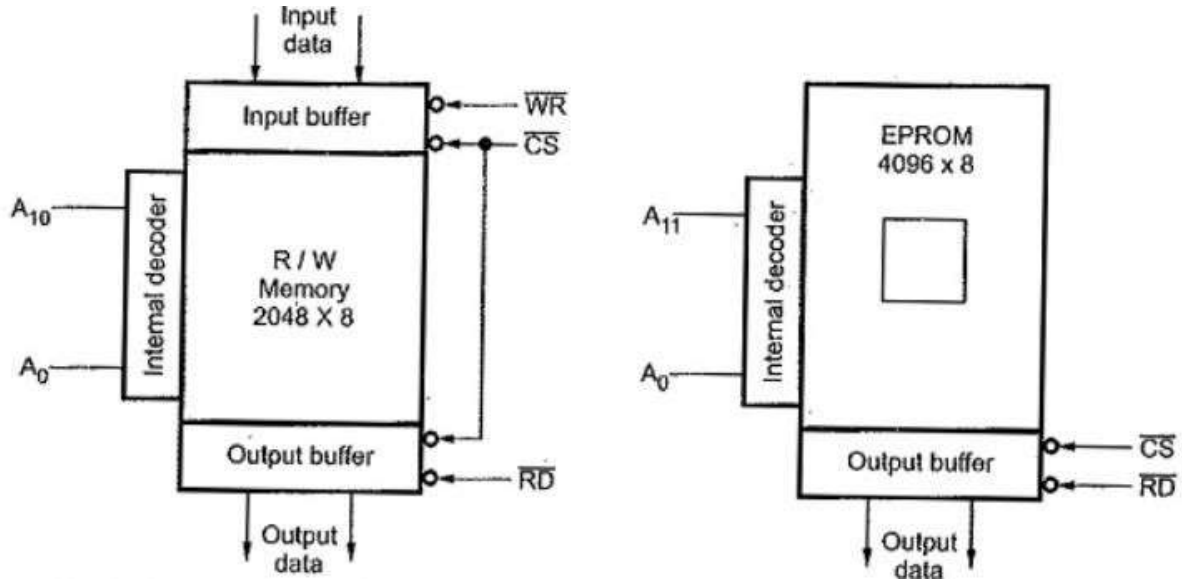
1. Memory Interfacing:

Memory is an integral part of a microprocessor system, and in this section, we will discuss how to interface a memory device with the microprocessor. The Memory Interfacing in 8085 is used to access memory quite frequently to read instruction codes and data stored in memory. This read/write operations are monitored by control signals. The microprocessor activates these signals when it wants to read from and write into memory.

In the last section we have already seen the memory read and memory write machine cycles, and status of the RD, WR and IO/M status signals for read/write operation. In the following section we will see memory structure and its requirements, concepts in Memory Interfacing in 8085 and interfacing examples.

Memory Structure and its Requirements

As mentioned earlier, read/write memories consist of an array of registers, in which each register has unique address. The size of the memory is N x M as shown in Figure below where N is the number of registers and M is the word length, in number of bits.



Basic Concepts in Memory Interfacing:

For Memory Interfacing in 8085, following important points are to be kept in mind.

1. Microprocessor 8085 can access 64Kbytes memory since address bus is 16-bit. But it is not always necessary to use full 64Kbytes address space. The total memory size depends upon the application.

2. Generally EPROM (or EPROMs) is used as a program memory and RAM (or RAMs) as a data memory. When both, EPROM and RAM are used, the total address space 64Kbytes is shared by them.

3. The capacity of program memory and data memory depends on the application.

4. It is not always necessary to select 1 EPROM and 1 RAM. We can have multiple EPROMs and multiple RAMs as per the requirement of application.

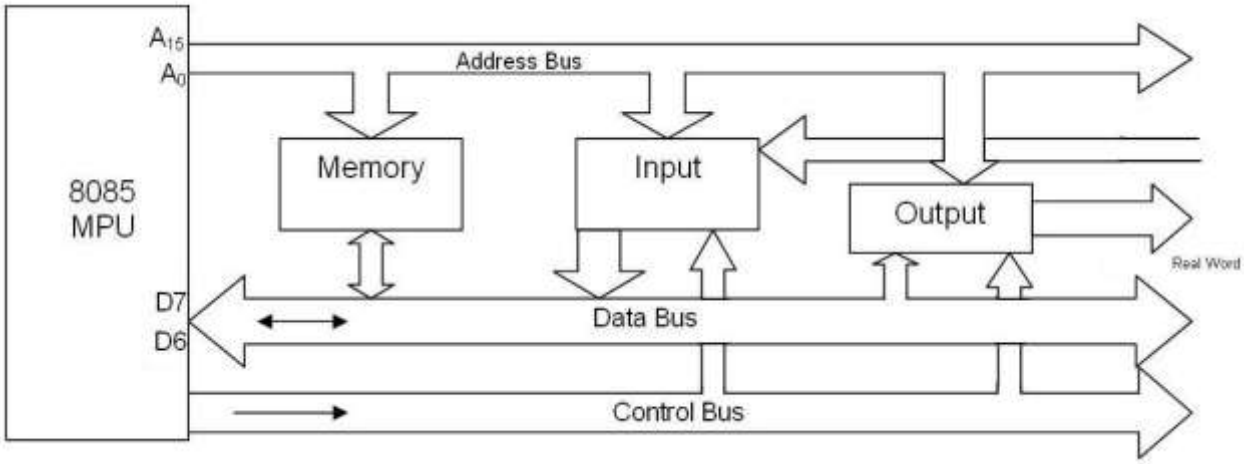
5. We can place EPROM/RAM anywhere in full 64 Kbytes address space

But program memory (EPROM) should be located from address 0000H since reset address of 8085 microprocessor is 0000H.

6. It is not always necessary to locate EPROM and RAM in consecutive memory For example: If the mapping of EPROM is from 0000H to OFFFH, it is not must to locate RAM from 1000H. We can locate itanywhere between 1000H and FFFFH. Where to locate memory component totally depends on the application.

Bus System

Lecture-4



There are three buses in Microprocessor:

1. **Address Bus**

2. Data Bus

3. Control Bus

1. **Address Bus:- Generally**, the Microprocessor has a 16-bit address bus. The bus over which the CPU sends out the address of the memory location is known as the Address bus. The address bus carries the address of the memory location to be written or to be read from.

The address bus is unidirectional. It means bits flowing occurs only in one direction, only from microprocessor to peripheral devices.

We can find how much memory location it can use the formula 2^N. where N is the number of bits used for address lines.

here, 2^16 = 65536bytes or 64Kb

So we can say that it can access up to 64 kb memory location.

Q.>If a processor has 4 GB memory then how many address lines are required to access this memory?
Ans: 4GB= 4 * 1GB
4 = 2^2

1GB = 2^30

4GB = 2^2 * 2^30 = 2^32

So 32 address lines are required to access the 4 GB memory.

2. Data Bus:- 8085 Microprocessor has an 8-bit data bus. So it can be used to carry the 8-bit data starting from 00000000H(00H) to 11111111H(FFH). Here 'H' tells the Hexadecimal Number. It is bidirectional. These lines are used for data flowing in both directions means data can be transferred or received through these lines. The data bus also connects the I/O ports and CPU. The largest number that can appear on the data bus is 11111111.
It has 8 parallel lines of data bus. So it can access up to 2^8 = 256 data bus lines.

3. Control Bus:- The control bus is used for sending control signals to the memory and I/O devices. The CPU sends a control signal on the control bus to enable the outputs of addressed memory devices or I/O port devices.
Some of the control bus signals are as follows:

1. Memory read
2. Memory write

3.I/O read

4.I/O writ

Pin Diagram & Functions

Lecture-5

Figure:8085PinDiagram

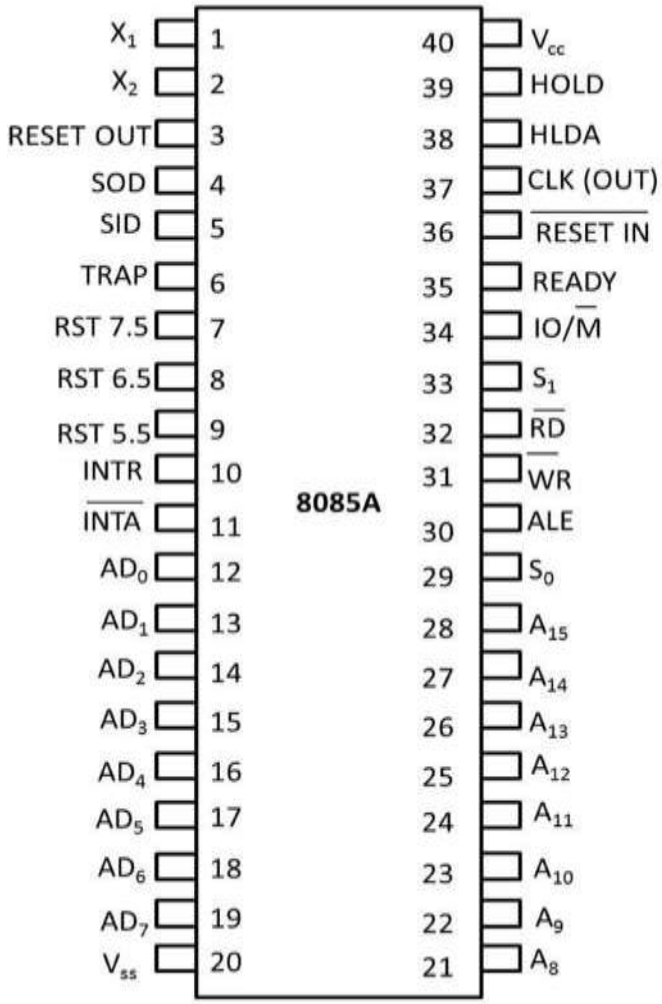


Figure: 8085 Pin Diagram

Allsignalscanbeclassifiedintosixgroups:

1. AddressBus
2. DataBus
3. Control&StatusSignals
4. PowerSupply&Frequencysignals
5. Externallyinitiatedsignals

6. SerialI/OPorts

1) AddressBus (pin 12to28)

- 16signallinesareusedasaddressbus.
- Howevertheselinesaresplitintotwosegments:A_{15}-A_8andAD_7-AD_0
- A_{15}-A_8areunidirectionalandareusedtocarryhigh-orderaddressof16-bitaddress.
- AD_7 -AD_0areusedfordualpurpose.

2) DataBus/Multiplexed Address (pin 12to19)

- SignallinesAD7-AD0arebidirectionalandservedualpurpose.
- Theyareusedaslow-orderaddressbusaswellasdatabus.
- Theloworderaddressbuscanbeseparatefromthesesignalsbyusingalatch.

3) Control&StatusSignals

- Toidentifynatureofoperation
- TwoControlSignals
  1) RD'(Read-pin32)
     - ✓ Thisisareadcontrolsignal(activelow)
     - ✓ ThissignalindicatesthattheselectedI/OorMemorydeviceistoberead&dataareavailableondatabus.
  2) WR'(Write-pin31)
     - ✓ Thisisawritecontrolsignal(activelow)
     - ✓ ThissignalindicatesthattheselectedI/OorMemorydeviceistobewrite.
- ThreeS tatusSignals
  1) S_1 (pin 33)
  2)S_0(pin29)
     - ✓ S_1andS_0statussignalscanidentifyvariousoperations,buttheyarerarelyusedinsmall systems.
     - ✓

S_1	S_0	Mode
0	0	HLT
0	1	WRITE
1	0	READ
1	1	OPCODEFETCH

3) IO/M’(pin34)
   - ✓ ThisisastatussignalusedtodifferentiateI/Oandmemoryoperation

- ✓ Whenitishigh,itindicatesanI/Ooperation
- ✓ Whenitislow,itindicatesamemoryoperation
- ✓ ThissignaliscombinedwithRD'andWR'togenerateI/O&memorycontrolsignals

- Toindicatebeginningofoperation
  - OneSpecialSignalcalledALE(AddressLatchEnable-Pin 30)
  - Thisispositivegoingpulsegeneratedeverytimethe8085beginsanoperation(machinecycle)
  - ItindicatesthatthebitsonAD7-AD0areaddressbits
  - Thissignalisusedprimarilytolatchthelow-addressfrommultiplexedbus&generateaseparatesetofaddresslinesA7-A0.

4) PowerSupply&FrequencySignal

- V_{cc}Pinno.40,+5VSupply
- V_{ss}Pinno.20,GroundReference
- X1,X2Pinno.1&2,CrystalOscillatorisconnectedatthesetwopins.Thefrequencyisinternallydividedbytwo;Therefore,to operate a system at3MHz,thecrystalshouldhaveafrequencyof6MHz.
- CLK(OUT) □ Clockoutput.PinNo.37:Thissignalcanbeusedasthesystemclockforotherdevices.

5) ExternallyInitiatedSignalsincludingInterrupts

- INTR(Input) □InterruptRequest.Itisusedasgeneralpurposeinterrupt
- INTA'(Output) □InterruptAcknowledge.Itisusedtoacknowledgeaninterrupt.
- RST7.5,RST6.5,RST5.5(Input) □RestartInterrupts.
  - Thesearevectorinterruptsthattransfertheprogramcontroltospecificmemorylocations.
  - TheyhavehigherprioritiesthanINTRinterrupt.
  - Amongthese3interrupts,thepriorityorderisRST7.5,RST6.5,RST5.5
- TRAP(Input) □Thisisanonmaskableinterrupt&hasthehighestpriority.
- HOLD(Input) □ ThissignalindicatesthataperipheralsuchasDMAControllerisrequestingtheuseofaddress&databuses
- HLDA(Output) □HoldAcknowledge.ThissignalacknowledgestheHOLDrequest
- READY (Input)This signal is used to delay the microprocessor read or writecycles until as low- responding peripheral is ready to send or accept data. Whenthe signal goes low, the microprocessor waits for an integral no. of clock cyclesuntilitgoeshigh.
- RESETIN'(Input)Whenthesignalonthispingoeslow,theProgramCounterissettozero,thebusesaretri-stated&microprocessorisreset.
- RESETOUT(Output) This signalindicatesthatmicroprocessorisbeing

reset.Thesignalcanbeusedtoresetotherdevices.

6) SerialI/O Ports

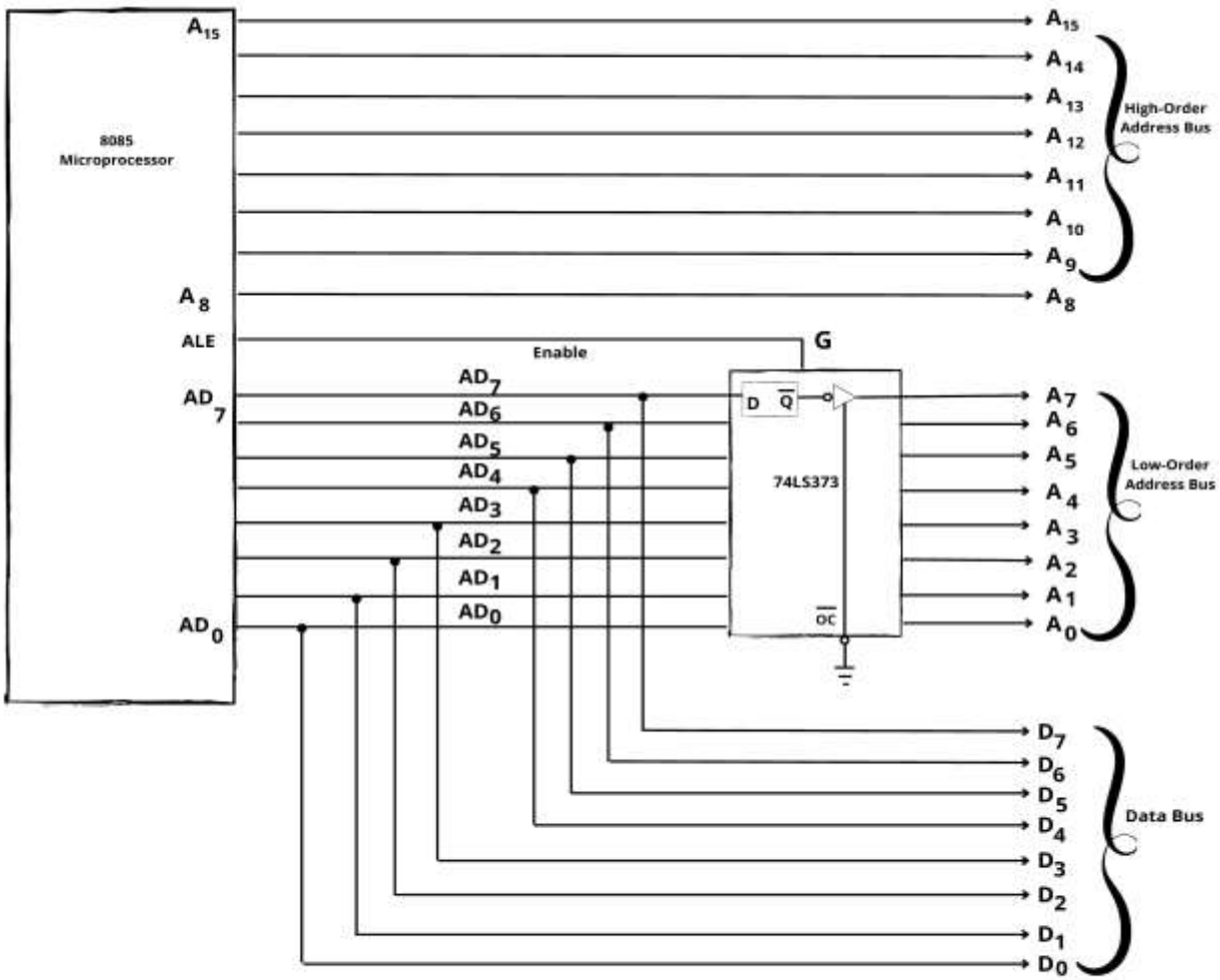
- Twopinsforserialtransmission
  1) SID(SerialInputData-pin5)
  2) SOD(SerialOutputData-pin4)
- Inserialtransmission,databitsaresentoverasingleline,onebitatatime.

Multiplexing & Demultiplexing of Buses

Lecture-6

For economic use of pins of **microprocessor 8085**, the lower order address bus is multiplexed with the data bus. this is known as **multiplexed Address/Data Bus (AD0 – AD7)**. the microprocessor 8085 provides an address during the earlier part of the execution of an instruction on the multiplexed bus (AD0 – AD7). data is also available on the same bus during the later part of the execution of an instruction. but for proper memory or input/output operation, the microprocessor requires the lower order address as well as data separately.

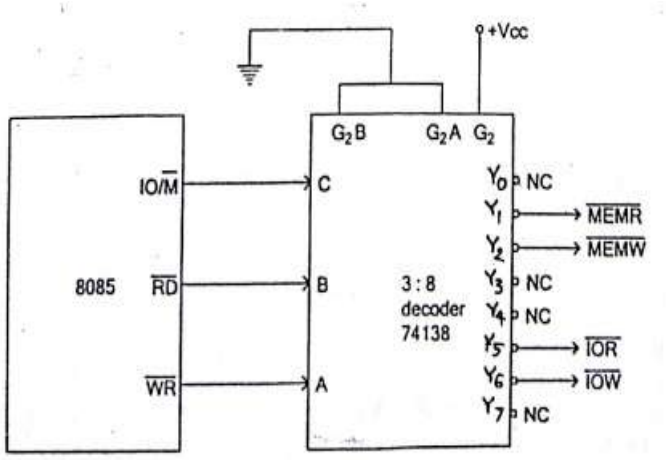
Address latch enables (ALE) signal is used to demultiplex the address/data bus. the below figure shows the schematic of demultiplexing of address/data bus (AD0 – AD7) using a latch IC (74LS373). the multiplexed address/data bus (AD0 – AD7) is connected as the input to the 8-bit latch IC (74LS373). the enable input (G) of latch IC is connected to the ALE pin of the microprocessor whereas the output control (OC) pin is grounded.


8085
Microprocessor
A15
A8
ALE
AD7
AD0
Enable
G
AD7
AD6
AD5
AD4
AD3
AD2
AD1
AD0
D
Q
74LS373
OC
A15
A14
A13
A12
A11
A10
A9
A8
High-Order
Address Bus
A7
A6
A5
A4
A3
A2
A1
A0
Low-Order
Address Bus
D7
D6
D5
D4
D3
D2
D1
D0
Data Bus

Generation of Control Signals

Lecture-7

Functions of Control Pins: IO/M' – It is a status signal which determines whether the address is for input-output or memory. When it is high (Logic 1) the address on the address bus is for input-output devices. When it is low (Logic 0) the address on the address bus is for the memory. • RD' – It is a signal to control READ operation. When it is low the selected memory or input-output device is read. • WR' – It is a signal to control WRITE operation. When it goes low the data on the data bus is written into the selected memory or I/O location.



3:8 Decoder A 3 to 8 line decoder IC 74138 is used. It has 3-inputs and 8-outputs. Three signals from the 8085 microprocessor namely read (RD'), write (WR') and Input Output/Memory (IO/M') are connected as input to the decoder. Four output signals are taken from output pins of decoder.

Control Signals The table list input conditions required or different inputs. • S1, S0 – These are status signals. They distinguish the various types of operations such as halt, reading, instructions fetching or writing. • RD', WR' – Read and write are two basic control signals for reading and writing operations respectively.

Functions of the control signals

• MEMR' – It indicates Memory Read operation • MEMW' – It indicates Memory Write operation • IOR' – It indicates Input Read operation • IOW' – It indicates Output Write operation

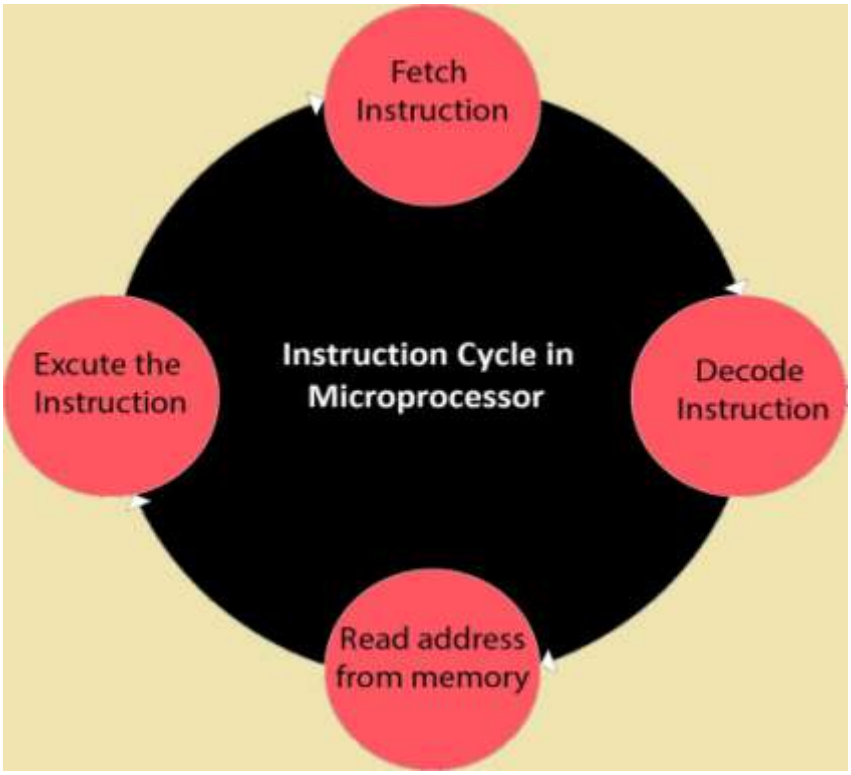
Instruction Cycle, Machine Cycle & T State

Lecture-8

Instruction Cycle in Microprocessor

The time required for the completion or execution of one instruction is known as the Instruction Cycle in Microprocessor. The instruction cycle consists of a –

- Fetch Instruction Cycle in Microprocessor
- Decoding Instruction Cycle in Microprocessor
- Reading effective address Instruction Cycle in Microprocessor
- Execution Instruction Cycle in Microprocessor



Fetch Instruction Cycle in Microprocessor

The next instruction is fetched by the address stored in the program counter (PC) and then stored in the instruction register.

Decode Instruction Cycle in Microprocessor

Decoder interprets the encoded instruction from the instruction register.

Reading effective address Instruction Cycle in Microprocessor

The address given in the instruction is read from the main memory and the required data is fetched. It depends on the direct addressing mode or the indirect addressing mode.

Execution Instruction Cycle in Microprocessor

It consists of the memory read (MR), memory write (MW), input-output read (IOR) and input-output write (IOW).

Machine Cycles in Microprocessor

- The time which is required for accessing the memory or input/output devices is called a machine cycle in Microprocessor.
- One to four machine cycles are used.
- For example, memory reading, memory writing, opcode fetches, etc.
- One machine cycle is equal to 12 clocks.
- One machine cycle consists of 6 states.

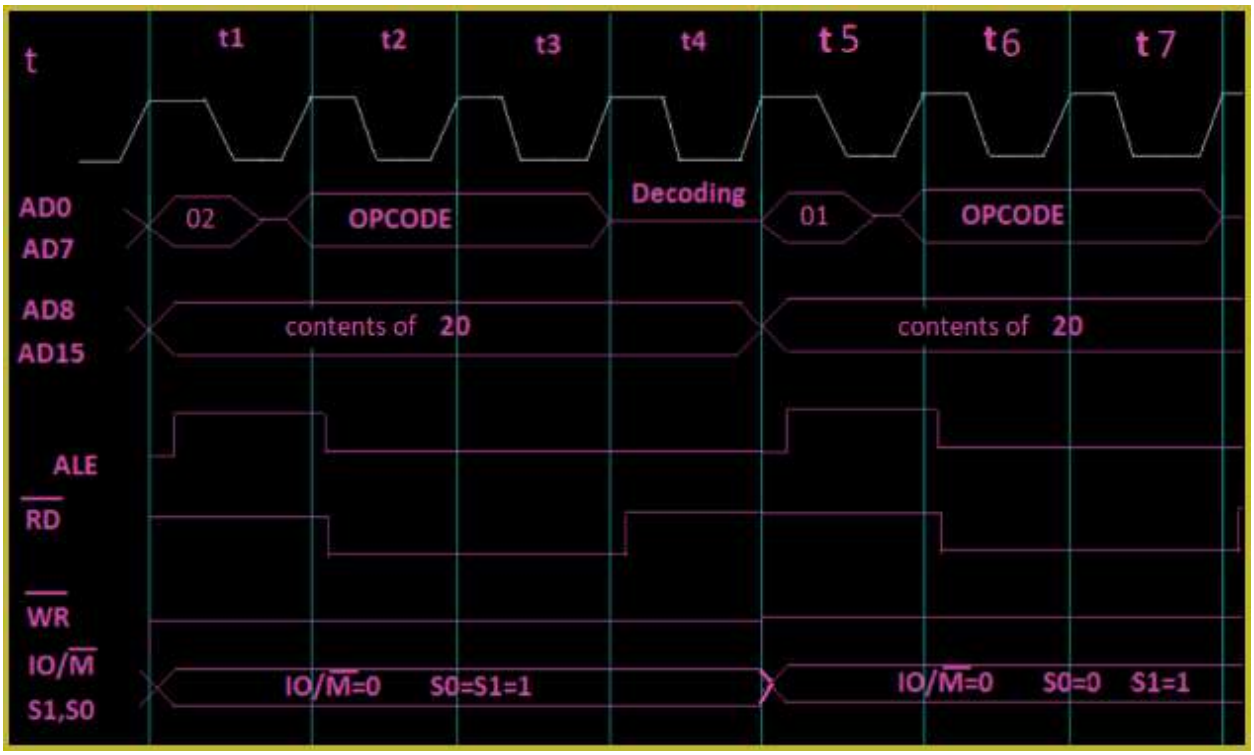
T states in Microprocessor

- It is one subdivision of the operation performed in one clock period.
- It is part of the machine cycle.

- One clock period is equal to one t state.
- One machine cycle is equal to 12 t states.

Timing Diagram in Microprocessor

The timing diagram in Microprocessor is a graphical representation. It is used for the representation of the execution time taken by each instruction in a graphical format. The t states represent the timing diagrams.



1. **CLOCK** – It is the clock pulse provided to the user.
2. **AD0-AD7** – These are used to carry data and addresses. It is the lower address bus. The decision of when it will carry an address and when it will carry data is made by ALE.
3. **A8-A15**– It is used to carry the address bits.
4. **ALE** – It provides the signal for multiplexed addresses and data buses. If the signal is high or 1, a multiplexed address and data bus will be

used as an address bus. To fetch a lower bit of address, the signal is 1 so that the multiplexed bus can act as an address bus. If the signal is low or 0, the multiplexed bus will be used as the data bus. When a lower bit of address is fetched then it will act as a data bus as the signal is low.

5. **RD (low active)** – If the signal is high or 1, no data is read by the microprocessor. If the signal is low or 0, data is read by the microprocessor.
6. **WR (low active)** – If the signal is high or 1, no data is written by the microprocessor. If the signal is low or 0, data is written by the microprocessor.
7. **IO/M (low active) and S1, S0** – If the signal is high or 1, the operation is performed on input-output. If the signal is low or 0, the operation is performed on memory.